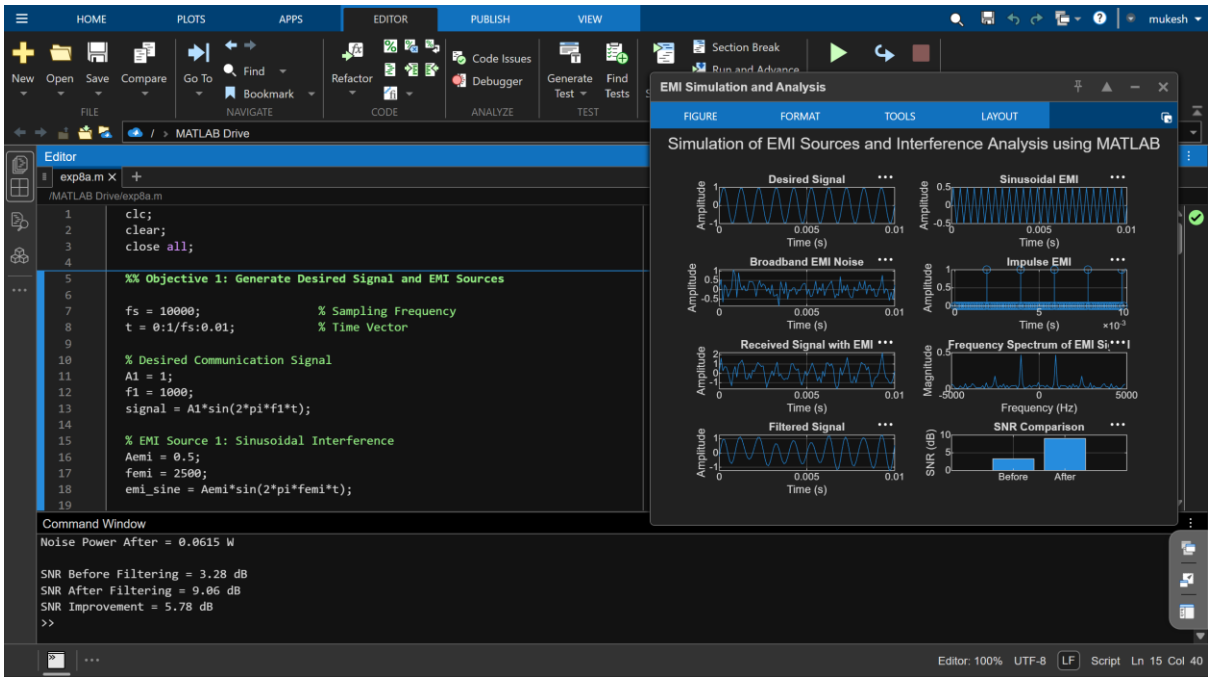
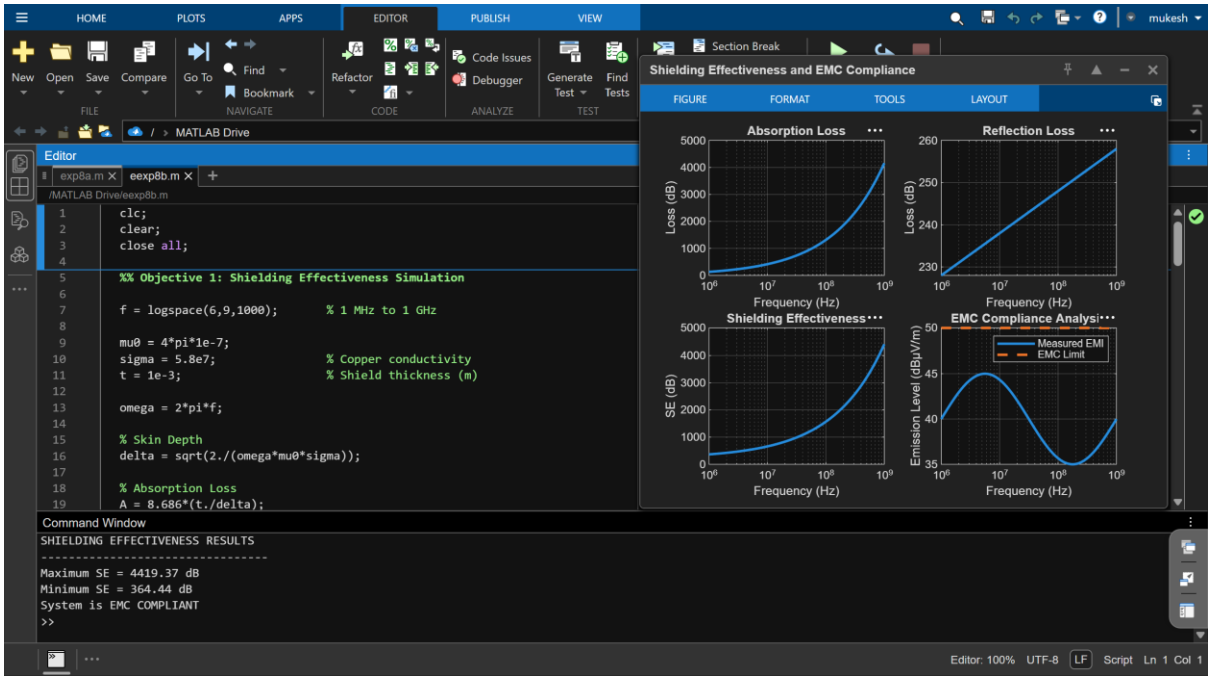
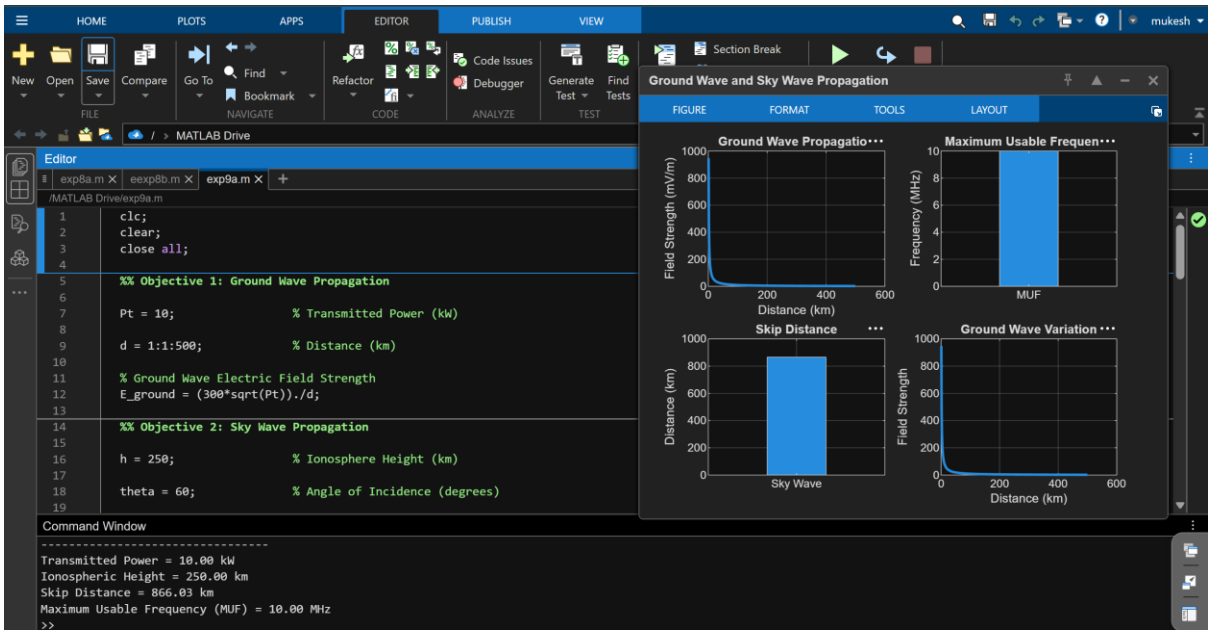
Exp8.1


EMI Simulation and Analysis
Simulation of EMI Sources and Interference Analysis using MATLAB
Desired Signal
Sinusoidal EMI
Broadband EMI Noise
Impulse EMI
Received Signal with EMI
Frequency Spectrum of EMI Si
Filtered Signal
SNR Comparison
Time (s)
Frequency (Hz)
Amplitude
Magnitude
SNR (dB)
Before
After
clc;
clear;
close all;
%% Objective 1: Generate Desired Signal and EMI Sources
fs = 10000; % Sampling Frequency
t = 0:1/fs:0.01; % Time Vector
% Desired Communication Signal
A1 = 1;
f1 = 1000;
signal = A1*sin(2*pi*f1*t);
% EMI Source 1: Sinusoidal Interference
Aemi = 0.5;
femi = 2500;
emi_sine = Aemi*sin(2*pi*femi*t);
Command Window
Noise Power After = 0.0615 W
SNR Before Filtering = 3.28 dB
SNR After Filtering = 9.06 dB
SNR Improvement = 5.78 dB
>>

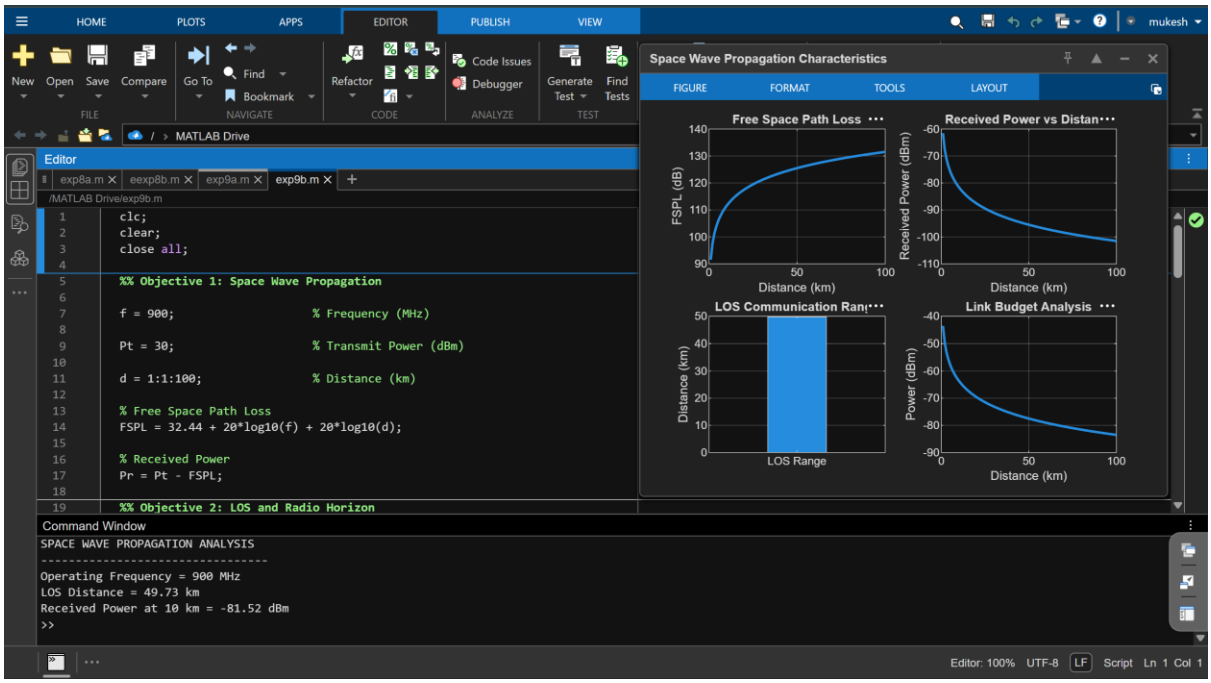
Exp8.2


Shielding Effectiveness and EMC Compliance
Absorption Loss
Reflection Loss
Shielding Effectiveness
EMC Compliance Analysi
Measured EMI
EMC Limit
Loss (dB)
SE (dB)
Emission Level (dBµV/m)
Frequency (Hz)
clc;
clear;
close all;
%% Objective 1: Shielding Effectiveness Simulation
f = logspace(6,9,1000); % 1 MHz to 1 GHz
mu0 = 4*pi*1e-7;
sigma = 5.8e7; % Copper conductivity
t = 1e-3; % Shield thickness (m)
omega = 2*pi*f;
% Skin Depth
delta = sqrt(2./(omega*mu0*sigma));
% Absorption Loss
A = 8.686*(t./delta);
Command Window
SHIELDING EFFECTIVENESS RESULTS
Maximum SE = 4419.37 dB
Minimum SE = 364.44 dB
System is EMC COMPLIANT
>>

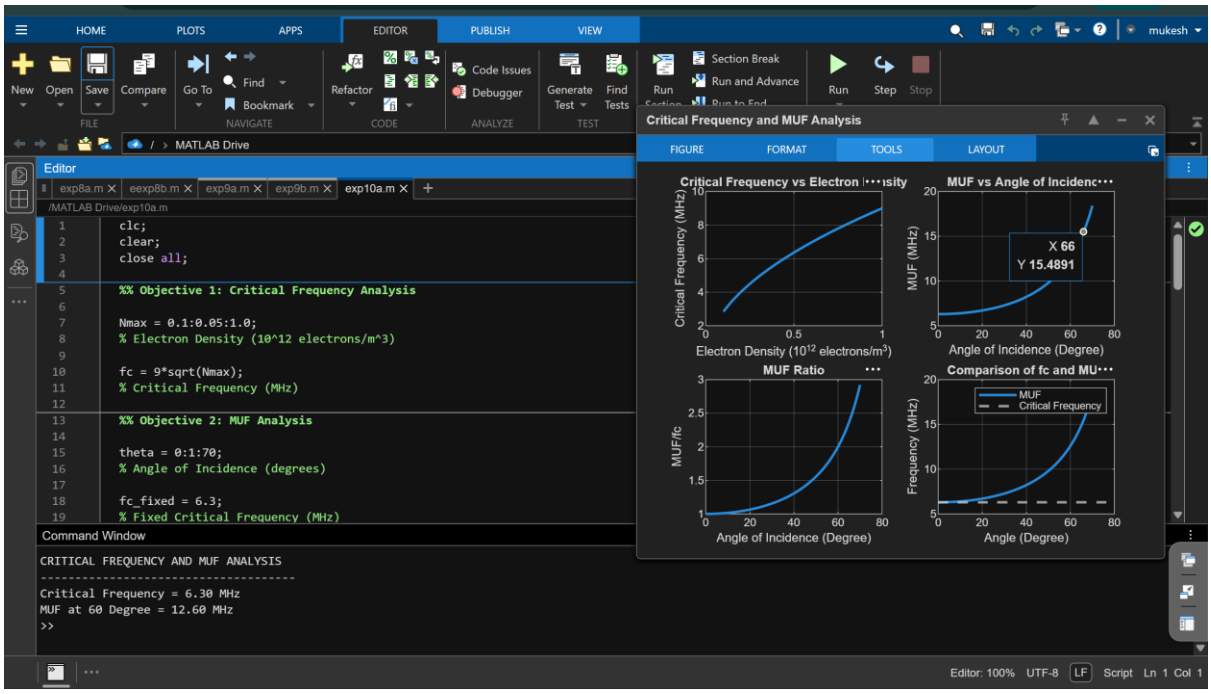
Exp9.1


Ground Wave and Sky Wave Propagation
Ground Wave Propagation
Maximum Usable Frequency
Skip Distance
Ground Wave Variation
Field Strength (mV/m)
Frequency (MHz)
Distance (km)
Sky Wave
MUF
clc;
clear;
close all;
%% Objective 1: Ground Wave Propagation
Pt = 10; % Transmitted Power (kW)
d = 1:1:500; % Distance (km)
% Ground Wave Electric Field Strength
E_ground = (300*sqrt(Pt))./d;
%% Objective 2: Sky Wave Propagation
h = 250; % Ionosphere Height (km)
theta = 60; % Angle of Incidence (degrees)
Transmitted Power = 10.00 kW
Ionospheric Height = 250.00 km
Skip Distance = 866.03 km
Maximum Usable Frequency (MUF) = 10.00 MHz

Exp9.2


Space Wave Propagation Characteristics
Free Space Path Loss
Received Power vs Distance
LOS Communication Range
Link Budget Analysis
FSPL (dB)
Received Power (dBm)
Distance (km)
LOS Range
Power (dBm)
clc;
clear;
close all;
%% Objective 1: Space Wave Propagation
f = 900; % Frequency (MHz)
Pt = 30; % Transmit Power (dBm)
d = 1:1:100; % Distance (km)
% Free Space Path Loss
FSPL = 32.44 + 20*log10(f) + 20*log10(d);
% Received Power
Pr = Pt - FSPL;
%% Objective 2: LOS and Radio Horizon
SPACE WAVE PROPAGATION ANALYSIS
Operating Frequency = 900 MHz
LOS Distance = 49.73 km
Received Power at 10 km = -81.52 dBm

Exp10.1



Exp10.2

